PROJECT BRIEF FOR JOURNAL PUBLICATION

Satya-Score: An AI Hallucination Detection & Scoring Framework

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At a glance	
Research area	Natural language processing · hallucination detection · fact verification · Indian languages
Languages	English and Hindi (Marathi planned)
What it does	Splits an AI chatbot answer into factual claims and marks each one Verified, Uncertain or Hallucinated, with its evidence and, when a structured source decided it, the correct fact
Evidence sources	World Bank and WHO statistics · Wikidata knowledge graph · professional fact-checks · 102,069 English and Hindi Wikipedia passages
Data	456 human-labelled chatbot claims (266 English, 190 Hindi) + 40 claims from two further LLMs
Status	Fully implemented web system; 265 automated tests; copyright filing through the institute in progress
Suggested paper type	Full research article (system + evaluation), or a resource/system paper

1. Summary

Large language models (LLMs) write fluent answers that sometimes contain false statements, known as hallucinations. Readers cannot easily tell which sentence is wrong, and checking tools for Indian languages are scarce. Satya-Score is a claim-level hallucination detection and scoring framework for English and Hindi. It divides an answer into individual factual claims and routes each claim to the most dependable source available: official statistics from the World Bank and the WHO and the Wikidata knowledge graph first, then published professional fact-checks, and otherwise Wikipedia passages judged by a multilingual natural-language-inference (NLI) model. Every claim receives a verdict, a score and its evidence, and, when a structured source decided it, the correct fact. On real chatbot claims the system rarely flags true statements (about 3% false alarms on unseen true text), and structured sources decide almost without error; hallucinations that no structured source covers remain hard to catch from encyclopaedic text alone, and we report this openly.

2. Why the problem matters

- Chatbots are used for health, education, news and public information; an unnoticed false fact can mislead many people.
- Errors are usually local: one wrong date or name in an otherwise correct answer. A checker must work claim by claim and show its evidence.

- Most hallucination detectors are English-only. Hindi has hundreds of millions of speakers but few tools; the strongest multilingual systems rely on very large LLMs.

3. What is new (contributions)

- **Source routing with explanations.** Each claim goes to the most authoritative source that covers it, and the verdict is explained with that source and, when a structured source decides, the correct fact (e.g. “Correct fact: Mahatma Gandhi · place of birth: Porbandar”).
- **Bilingual structured verification.** Wikidata checks for places, dates, capitals, authors, office-holders, quantities and currencies with English and Hindi templates; a namesake rule (any matching entity supports; a contradiction needs every namesake to disagree).
- **Two-agency statistics rule.** Country statistics are checked against the World Bank and, for life expectancy and infant mortality, also the WHO; when both report a figure they must agree, and figures are compared at the precision the claim is written to.
- **Negation-safe fact-check matching.** A published “False” rating applies only if the claim entails the checked claim, a “True” rating only in the other direction, and never on contradiction, so “hot water does not kill the virus” is not marked false by a fact-check of “hot water kills the virus”.
- **Conservative evidence rule.** Support may come from any retrieved passage, but only a passage about the claim’s own subject may contradict it; this cut false alarms on fresh true sentences from 33 to 8 of 99, but also lowered hallucinations caught on training claims from 11 to 3 of 45 and failed one pre-fixed check, so it was adopted as a stated product decision.
- **A human-labelled bilingual dataset and a disciplined evaluation.** 456 English and Hindi chatbot claims, labelled by a team of four annotators reviewing AI suggestions and checked in two blind rounds; most design changes were tested against a decision rule fixed in advance, and the four decisions that went against a rule's outcome are reported, so the paper can report negative results honestly.

Novelty in one sentence: earlier work treats claim-level checking, multilingual (Hindi) detection and knowledge-graph grounding separately; Satya-Score combines them, together with official statistics and professional fact-checks, in one explainable, near real-time system. The individual models and data sources are existing public resources; the contribution is the combination, the decision rules and the evaluation.

4. How it works

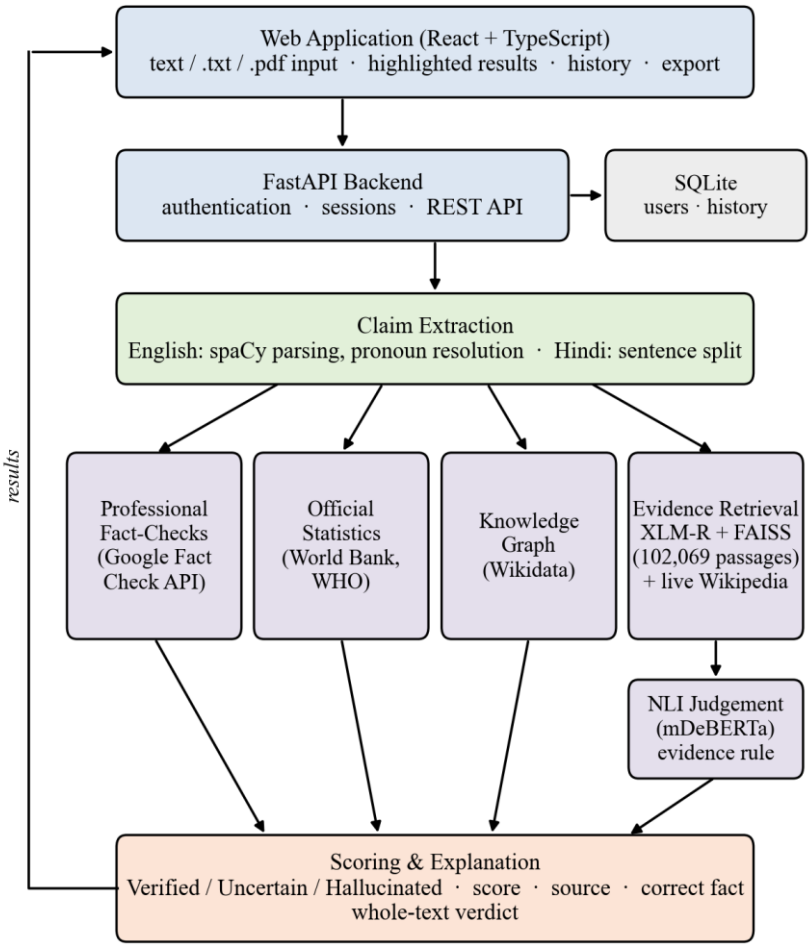


Figure 1. Architecture of Satya-Score

- **Claim extraction:** English answers are parsed with spaCy and split into self-contained claims, with pronouns resolved to the sentence topic (correct referents on 400 held-out texts: 93 → 277); Hindi is split into sentences.
- **Retrieval:** multilingual XLM-R sentence embeddings and a FAISS index of 102,069 Wikipedia passages (89,594 English, 12,475 Hindi), extended on demand from live Wikipedia.
- **Judgement:** a multilingual mDeBERTa NLI model scores each passage; the evidence rule turns the scores into a verdict. A structured source that holds the fact decides directly.
- **Scoring (CMHSA):** a statistics or Wikidata answer sets the claim score (1.0 or 0.0), otherwise a matching fact-check (0.05 or 0.95), otherwise the NLI evidence score; similarity has weight 0 because it measures topic, not truth. Below 0.35 is Hallucinated, above 0.65 Verified; a text is Hallucinated when at least a third of its claims are.
- **System:** FastAPI back end and React web application with accounts, history, export and a secure public link; a 6-claim answer takes about 5 s the first time and under 1 s when repeated on a laptop GPU.

5. Data and evaluation setup

- 456 claims from Llama 3.3 answers in five domains (government, health, science, sports, history): 266 English, 190 Hindi. Labelled by a team of four annotators (the authors), who shared the claims and reviewed AI-suggested verdicts; two annotators labelled a blind sample of 56 claims: binary agreement $\kappa = 0.53$ (88% raw).
- Grouped split (claims from one question stay together): 303 training, 76 test claims. 40 further relational claims from two other LLMs for the knowledge-graph study.
- Unseen true text: lead sentences of Wikipedia articles not used in development. Statistics: 40 randomly chosen countries. Fact-checks: 25 rumours and 15 true claims.

6. Main results

Table 1. Results of the live system

Test	Result
Unseen true Wikipedia sentences (128)	113 Verified; 4 false alarms
Training claims (303): true / hallucinated	9 of 258 false alarms; 5 of 45 hallucinations caught
Test claims (76): true / hallucinated	3 of 66 false alarms; 0 of 10 caught from Wikipedia text alone
Knowledge-graph decisions (40 relational claims, English)	11 of 13 correct (pre-registered); 12 of 12 after a linking fix (exploratory)
Knowledge-graph ablation, pre-registered (same claims)	AUC 0.704 → 0.784; Hallucinated-F1 0.512 → 0.615
Official statistics (40 randomly chosen countries)	137 of 137 false figures caught; no true figure contradicted
Professional fact-checks (25 rumours, 15 true claims)	0 wrong decisions; rumours caught 3 → 10; correct 13 → 20 of 40
Evidence rule (99 fresh true statements)	false alarms 33 → 8 (but training hallucinations caught 11 → 3 of 45)
Speed (6-claim answer, laptop GPU)	27.7 s → 5.1 s first time; 10.1 s → 0.7 s repeated; same verdicts

Reading of the results: when a structured source covers a claim, the verdict is almost always right and comes with the correct fact; true text rarely raises a false alarm. When only encyclopaedic text is available, hallucinations are often missed, because an NLI model can treat a passage on the same topic as support.

7. Changes tested and not adopted (negative results)

Table 2. Changes tested and decisions taken

Change	Outcome
Reading 10 passages instead of 3	correct training verdicts 130 → 170 of 303, but false alarms 8 → 11 of 258; adopted against the rule
Larger NLI models	favoured by the fixed rule, but gains within noise; current model kept
Number guard (two versions)	lost true Verified claims (115 → 101; 109 → 103)

Change	Outcome
MedlinePlus as a second evidence source	false alarms rose 9 → 11
Stricter support rule	true claims Verified fell 152 → 47
LLM judge (open mode)	caught 5 → 18 of 45 hallucinations but false alarms rose 9 → 35 of 258; met its rule, kept off by product decision
Grounding checker (MiniCheck, design probe)	still supported 7 of 9 false common-sense statements

Reporting these is a strength for a journal: it shows which plausible ideas do not help, with the evidence.

8. Limitations (stated in the paper)

- Hallucinations that no structured source covers (misconceptions, over-generalisations) are often missed; on 15 common-sense statements all 9 false ones were marked Verified.
- The labelled set is small (456 + 40 claims) and has few Hindi hallucinations; about 31 of the 45 training hallucinations came from an earlier version of the claim extractor rather than the chatbot itself. Each main claim was labelled once, by one of the four annotators, with the AI suggestion visible; independent agreement was moderate ($\kappa = 0.53$).
- The knowledge-graph study used English claims; Hindi checks are implemented but tested on fewer examples. The rule that the knowledge graph decides, and its linking fix, were chosen in exploratory analyses on the same claims.
- Live sources need internet access; WHO data is licensed for non-commercial use.

9. Position against related work

Table 3. Comparison with representative systems (as described in their papers)

System	Claim-level	Hindi	Structured sources	Shows correct fact
FActScore [2]	Yes	No	No (Wikipedia text)	No
SAFE [3]	Yes	No	No (web search + LLM)	No
SelfCheckGPT [4]	Sentence	No	No (no external source)	No
MiniCheck [5]	Sentence	No	No (given document)	No
Mu-SHROOM systems [6]	Span	Yes	No	No
MultiHal [7]	Benchmark	No	Knowledge-graph paths (evaluation resource)	Not applicable
Satya-Score	Yes	Yes	Wikidata, World Bank/WHO, fact-checks	Yes, when these decide

BHRAM-IL [8] benchmarks how often LLMs hallucinate in Hindi and other Indian languages; it evaluates models rather than checking a given answer, so it complements this work.

10. Implementation and availability

- Working web application (English/Hindi input, highlighted claims, sources, history, export), FastAPI back end, 265 automated tests.
- Can be shown live through a secure public link; packaged as a container for cloud hosting.
- Code and the labelled dataset: available to reviewers on request; public release planned after the copyright registration.

11. Possible journals

The work fits journals on natural language processing and low-resource languages. Examples to check (scope, indexing, fees and review time change, so verify on the journal's own website):

- ACM Transactions on Asian and Low-Resource Language Information Processing (TALLIP): Hindi and Indian-language focus.
- IEEE Access: broad, open access, system papers with solid evaluation.
- Natural Language Processing Journal (Elsevier) or Language Resources and Evaluation (Springer, for the dataset angle).
- SN Computer Science (Springer): applied systems with evaluation.

Caution: avoid journals that promise acceptance or publication within days for a fee; prefer journals indexed in Scopus or Web of Science, and check the institute's approved list (e.g. UGC-CARE) if publication must count officially.

Key references

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